The Art and Science of Gender-affirming Care for Clinicians

SASHA COHEN, MSN, FNP-C, AAHIVS

(PRONOUNS: THEY/HE)

OBJECTIVES

- Understand the dynamics of the Hypothalamic-Pituitary-Gonadal axis and how they are affected by exogenous hormone therapy
- Review fundamentals of prescribing gender-affirming hormone therapy (GAHT)
- Review laboratory monitoring of GAHT, recognize abnormal results and the recommended follow-up
- Discuss common issues experienced by TGNB patients on GAHT and how to address them

GAHT ASSESSMENT: Beyond Pronouns

As a clinician you hold a lot of power and influence over TGNB patient's experience of their transition.

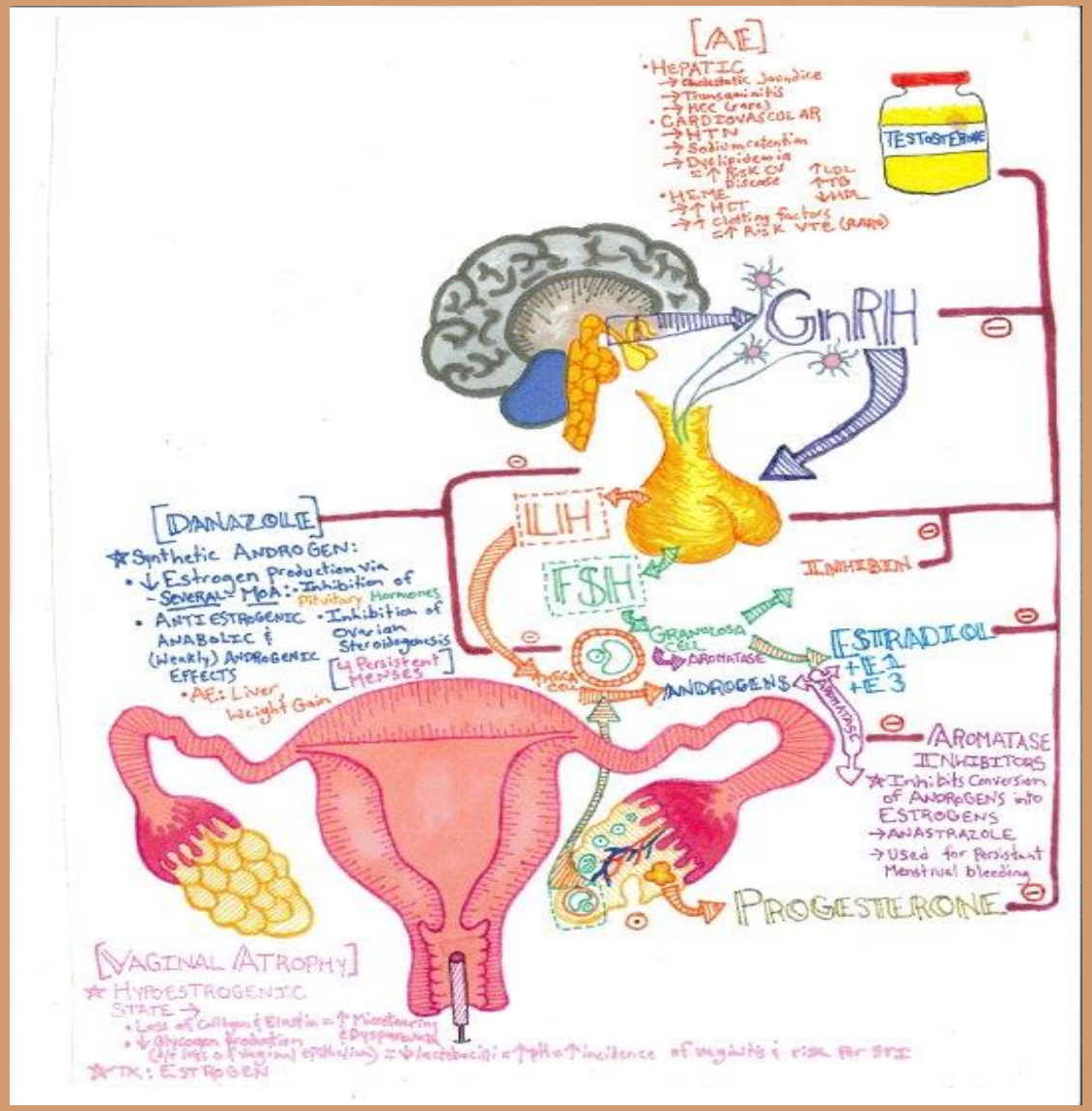
- Tips for creating an affirming experience for patients
  - Consider not only about **WHAT** you are saying but **HOW** you say it
  - Normalize sharing your pronouns
  - Mirror patient's terminology to the extent you feel comfortable
  - Communicate transparently with patients about the risks of GAHT but take care with how you discuss risk assessment/side effects
  - Let patients know that, while there are irreversible changes that result from GAHT there is also a lot of flexibility and ways to make their transition their own

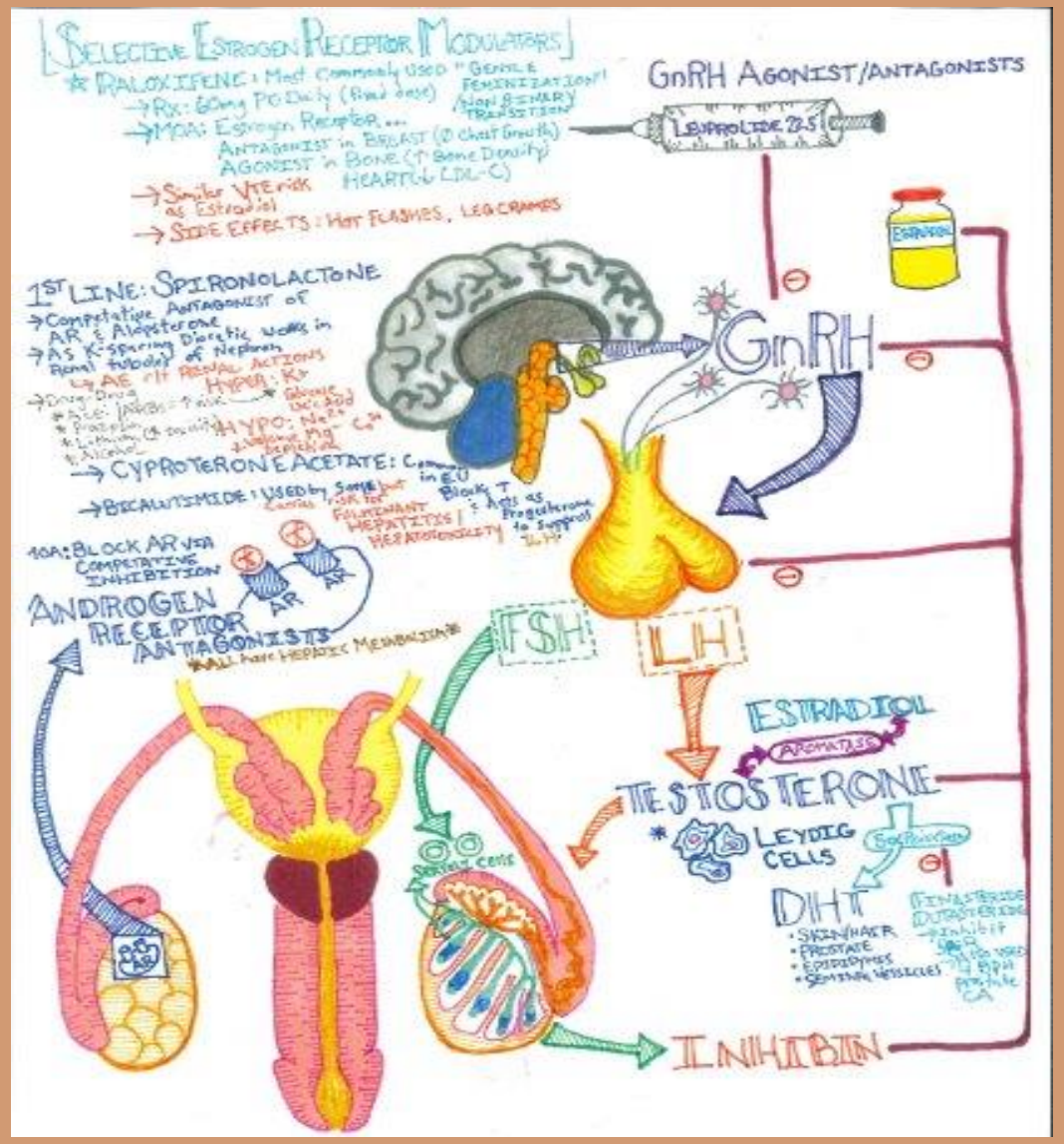
GAHT ASSESSMENT: Taking a Gender History

- Gender identity
- Preferred Pronouns
- GAHT patient goals
- GAHT patient concerns
- Social transition
- Social supports
- Legal name/gender marker change
- Family planning
- Binding/tucking
- Hx GAHT
- Hx gender affirming surgery
- Gender-affirming procedure goals

THE NUTS AND BOLTS OF PRESCRIBING GENDER-AFFIRMING HORMONE THERAPY

HYPOTHALAMIC-PITUITARY-GONADAL AXIS OVERVIEW



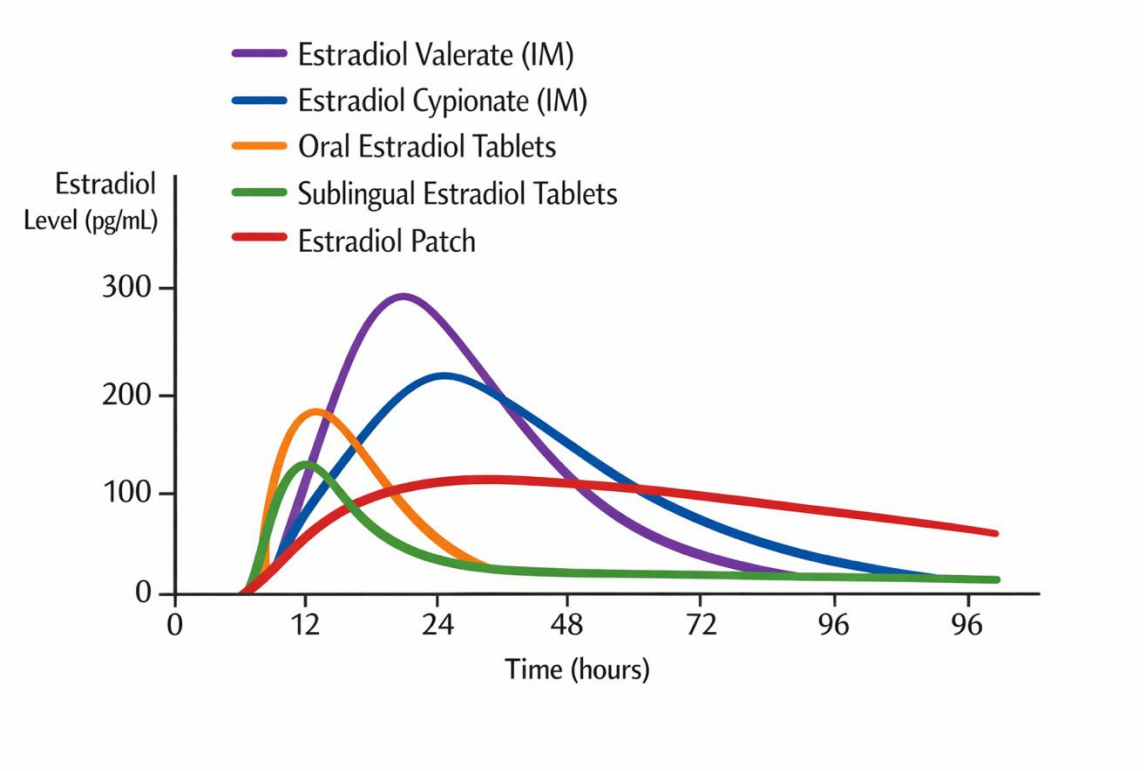


PRESCRIBING GAHT: General Guidance

- Bioidentical is best!
- Consider patient tolerance, safety & side effects
- Comorbid conditions/risk factors
- *Not a one-size fits all model*: helping patients choose the best regimen for them
  - Consider patient goals/priorities
    - Transition timeline/pace of changes
    - Modality preference
    - Sexual function
    - Fertility preservation?

PRESCRIBING GAHT: Feminizing Estradiol Regimens

- **Formulations & Pharmacokinetics**
  - Estradiol valerate
    - Bioavailability: 100% (IM/SC); ½ life: 4-5 days
  - Estradiol cypionate
    - Bioavailability: 100% (IM/SC); ½ life: 8-10 days
  - Oral
    - Bioavailability: 5%; ½ life: 13-20 hours
  - Sublingual
    - Bioavailability: 10%; ½ life: 8-18 hours
  - Transdermal
    - Bioavailability: 3-5%*; ½ life: ~37 hours



PRESCRIBING GAHT: Feminizing Regimens

Progesterone

There have been no well-designed studies of the role of progestogens in feminizing hormone regimens and its' use remains a controversial topic in TGNB medicine

- Many transfemine spectrum individuals and TGNB-care providers report an anecdotal improved breast and/or areolar development, positive effect on mood & sleep, and improvement in libido with the use of progestogens
- WPATH SOC 8 specifically does not recommend adding progesterone to GAHT regimen, but does not explicitly recommend against it
- Only approved use is for improved sexual function post-gonadectomy and as adjunct androgen blockade in addition to or instead of spironolactone, GnRHa

PRESCRIBING GAHT: Feminizing Regimens

Progesterone

- Safety
  - Bioidentical preferred (micronized)
  - Hepatic metabolism
  - Previously avoided d/t concern of increased breast cancer risk, which was based on data from the WHI and did not bear out in practice
- Clinical Logistics
  - Dosing: Prometrium 100-200mg qHS
    - Alternative: MPA 150mg IM q3mo
  - Consider waiting to initiate until at least 1 year on GAHT*
  - Lab monitoring not recommended

- **Patient Education**
  - Effects are highly variable
    - Not an essential part of GAHT regimen so if not seeing desired effects of experiencing side effects, fine to discontinue
    - Related to androgens so may cause virilization (rare)
    - No evidence for the efficacy of cycling progesterone

PRESCRIBING GAHT: Feminizing Regimens

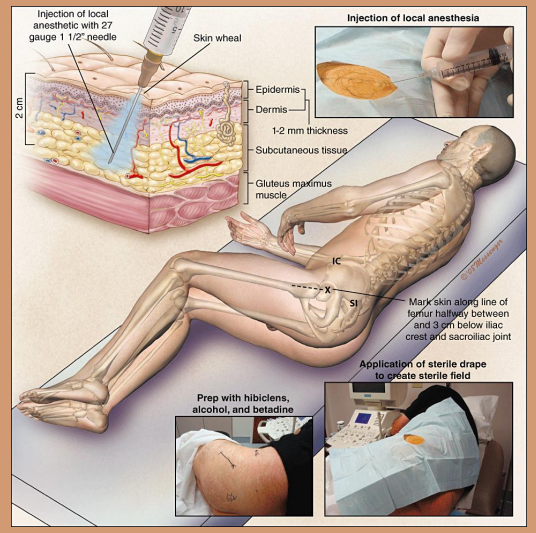
Antiandrogens

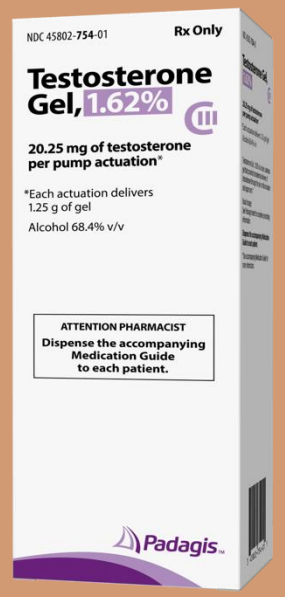
	Pharmacology	Benefits/Pros	Risks/Cons	Other Considerations
Spironolactone (Aldactone)	**-Steroidal -Binds directly to AR -Biovailability increases w/ adiposity**	**-cost-effective -most effective/well studied option for T suppression**	**-risk of hyperkalemia -diuretic effects**	**Dosing: 25mg-400mg daily in single or divided doses**
5-alpha-reductase Inhibitors (Finasteride, Dutasteride)	-prevents conversion of T into DHT -No effect on total testosterone	-decreases certain secondary sex characteristics (primarily androgenic alopecia and facial/body hair growth)	-not the most effective option for testosterone suppression if desired	Dosing: 1mg-5mg
GnRH agonists/antagonists (Lupron)	-direct suppression of endogenous production via GnRH receptor binding	-Very effective at suppressing endogenous hormone production	-expensive -in-clinic injections -must be used w/ cross-sex hormones for BMD	Dosing: 22.5mg q3 months
Cyproterone Acetate	-Steroidal -Binds directly to AR	-Blocks T and DHT very effectively	-Negative effect of serum lipids/ serious cardiovascular events -Hepatotoxicity	Dosing: 50mg-100mg daily Unavailable in US
Estradiol Monotherapy	-at adequate levels can suppress HPG axis via negative feedback	-No need to use additional medications=fewer side effects/less frequent dosing	-may not be possible for everyone to suppress T in this way	Dosing: higher doses or more frequent dosing may be needed

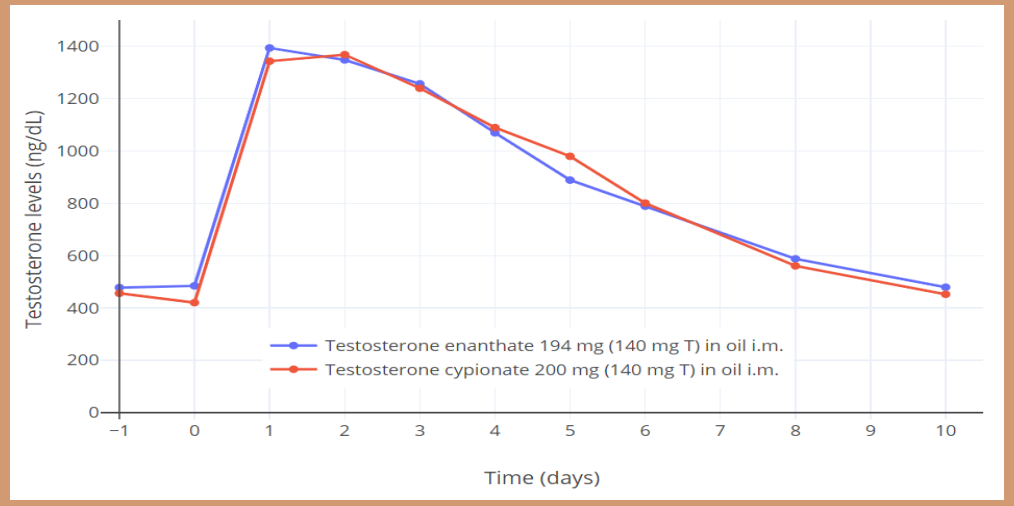
PRESCRIBING GAHT: Masculinizing Regimens

Testosterone

- Topical vs. Injections
  - Alternatives: Testopel, Aveed
- Dosing schedule: Daily vs. weekly vs. Biweekly
- Insurance coverage/cost
- Pt tolerance
- Patient goals/priorities
  - Transition timeline: the myth of the "microdosing"
  - Fertility preservation







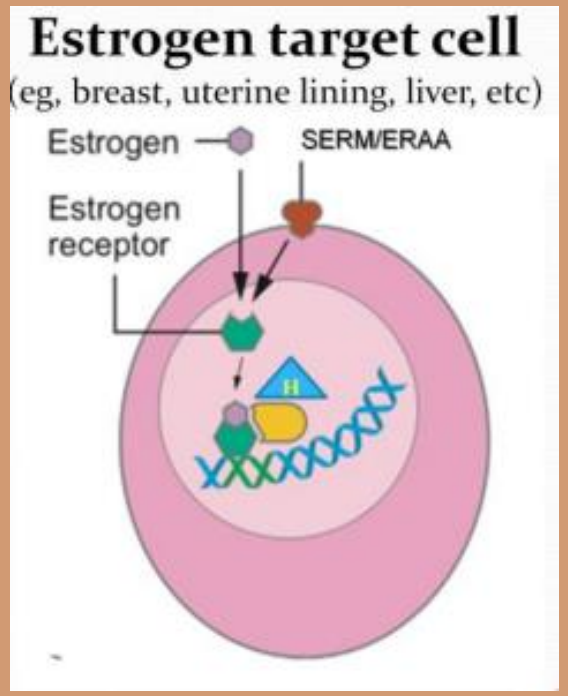
PRESCRIBING GAHT: Masculinizing Regimens

	Pharmacology	Dosing	Benefits/Pros	Risks/Cons	Other Considerations
Testosterone Cypionate	-Bioavailability ~95% -½ life: ~8 days -Pharmacokinetics similar w/ IM and SC dosing	Starting: 50mg weekly Max: 100mg weekly	-less frequent dosing (weekly or bimonthly) -Easier to suppress HPG axis d/t higher peak	-Injection site rxns/issues -Peak/trough fluctuations	-Cheapest option
Testosterone Enanthate	-slightly shorter ½ life than cypionate (~4.5 days)	Same as test cypionate	-as above -Good option for those w/ allergic rxns to cypionate*	-same as cypionate	-comes in 5ml vial
Topical Testosterone	-Bioavailability: ~8-14%	Starting:20-50mg daily Max: 101mg daily	-less fluctuation -possibly better for more gradual effect	-difficult to suppress HPG axis -transfer to others	-many formulations/brands may need PA Smell/texture may be off-putting
Testosterone undecanoate (Aveed, Jatenzo)	-Oral: low bioavailability; absorbed by GI tract into lymphatic system to avoid hepatic metabolism -Aveed: long ½ life (~21 days)	Jatenzo initial: 158mg BID Max: 316mg BID Aveed: 1000mg q12 weeks	-convenience of daily pill -Less frequent injection, administered by provider	risk of POE (Aveed) Hepatotoxicity? (oral) rapid hair loss? (Aveed) -Insurance coverage	-Aveed: facility and provider must complete training/be certified
Testosterone pellets (Testopel)	designed for consistent delivery; studies show levels peak around 1 month and then decrease (dissolution rate influences delivery)	2-10 75mg pellets, 150-750mg total. Every 3-4 months	-In-office procedure, pt does not need to self-administer -Great option for patients with scar tissue, executive fxn issues	Site reactions, pellet extrusion, infection	Expensive if not covered by insurance

PRESCRIBING GAHT: Alterative Regimens

Selective Estrogen Receptor Modulators (SERMs)

- MOA: interacts with intracellular estrogen receptors in target organs as estrogen receptor agonists or antagonists in a tissue-specific manner.
  - Agonist in Breast/Chest
  - Antagonist in Bone (⇧bone density) & Heart (⇩LDL-C)
- Typically used in infertility treatment, estrogen receptor–positive breast cancer, and prevention/treatment of osteoporosis in postmenopausal women.



PRESCRIBING GAHT: Alterative Regimens

Selective Estrogen Receptor Modulators (SERMs)

- Ideal option for those who desire a more non-binary presentation or who do not want chest tissue growth
  - May be used in conjunction with antiandrogen for increased feminizing effect
  - Ok to use w/o cross-sex hormone replacement d/t bone protective mechanism
  - Similar VTE risk to estradiol- **should NOT be combined with estradiol**
  - Possible side effects: hot flashes, leg cramps

Gender-affirming Hormone Therapy Lab Interpretation

Lab Interpretation: General Guidance

- It is expected for certain lab values to be within expected ranges for gender identity (rather than presumed sex)
  - Ex: H&H, Creatinine
- Be sure to counsel patients on what changes are expected so they are not concerned when they see all the red exclamation points on their labs
- Hormone level interpretation is an imperfect and often frustrating science. Hormone levels are a guidepost for you to assess safety and troubleshooting issues but should not be set in stone

Monitoring GAHT: When?

- **At Baseline:** CBC, CMP (+/-Lipids, TSH)
  - Assess for pre-existing pathology that may be affected by GAHT
  - No need to check baseline endogenous hormone levels as will not affect management
  - Provides opportunity to check other clinically indicted labs on an individual basis
  - Per UCSF guidelines no indication to monitor lipids while on GAHT*
- **Q3mo for 1st Year after GAHT Initiation or 3 Months After Dose/Administration Change:** CBC, CMP, Testosterone +/- Estradiol
  - Transfemme: CMP q3mo only if on spironolactone for androgen suppression
  - Follow-up sooner after GAHT initiation
- **Annually:** as above +/- Lipids

Monitoring GAHT: **Why?**

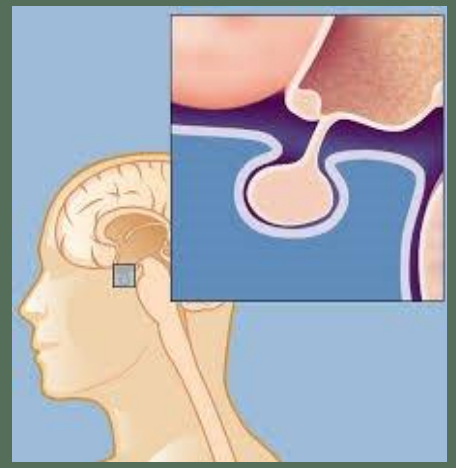
- Complete Blood Count
  - H&H, RBC: decrease in transfeminine patients; increase in transmasculine patients
    - **When to be worried:**
      - True anemia **always** needs to be worked-up
      - Polycythemia: HCT>52, increase monitoring, HCT>54 recommend therapeutic phlebotomy, HCT>56, refer to Heme
  - WBC: may see elevation in WBC w/ injectable estradiol- unclear why and conflicting data re: clinical significance
- Metabolic Panel
  - **BUN/Cr/K+**: if on spironolactone for testosterone-blocking
    - Ok to use expected "male" ULN for Cr in TGNB pts
    - **When to worry**
      - If K+ elevated it is always a cause for concern: assess for s/sxs of hyperkalemia & repeat lab to ensure not d/t hemolysis in lab sample. If persistently elevated – d/c spiro and repeat
      - Persistent/worsening renal function

Monitoring GAHT: Why?

- Metabolic Panel
  - Liver Function: May see slight increase in AST/ALT with initiation of testosterone in transmasculine spectrum patients, generally resolves spontaneously
    - Use caution if pre-existing liver disease
  - Guidelines for LFT monitoring for transfeminine spectrum people largely based on older studies using synthetic estrogens but still relevant, particularly with oral estradiol or supraphysiologic dosing
  - If persistently elevated, work-up alternate causes
- Lipids
  - Possible effects of Estradiol on lipids
    - Increase in TG and HDL, decrease in Total-C, LDL, ApoB
      - *if TG>1000 decrease estrogen dose until TG elevation addressed*
  - Possible effects of Testosterone on lipids
    - Studies have shown a clinically significant increase in LDL-C and a decrease in HDL-C, no clinically significant increase in TGs or TCHOL

Monitoring GAHT: Additional Labs When & Why

- Luteinizing Hormone (LH)
  - **Goal of GAHT is to achieve HPG axis suppression ----> should be eugonadal** (not producing endogenous hormones)
    - Check LH if e/o lack of HPG suppression(**if adequate suppression will be LOW**)
    - Can use to ensure adequate hormone levels post-gonadectomy (**if inadequate hormone replacement will be HIGH**)



Prolactin

- Prolactin secretion from anterior pituitary mediated by estradiol
  - C/f development of prolactinoma with supraphysiologic estradiol
- Routine monitoring previously recommended per accepted guidelines but this recommendation has been removed
  - Only check if s/sxs of prolactinoma (Vision changes, new onset headaches, excessive galactorrhea)
- If elevated + sxs: lower E2 dose and recheck in 3mo
- If persistent: Pituitary MRI

Monitoring GAHT: Hormone Levels

Measuring and interpreting hormone levels can be difficult as they fluctuate depending on method of administration and timing of labs

For patients on injectable GAHT: UCSF, WPATH SOC 8, & Endocrine Society guidelines all recommend mid-cycle measurement

Testosterone

- **Ideal mid-cycle range: ~400-800**
  - Primarily concerned about cardiovascular health and overall emotional stability w/ supraphysiologic levels
  - Rare risk for aromatization w/ persistent elevation
- It can be difficult to interpret testosterone levels in patients on topical testosterone- ask about timing of administration

Feminizing GAHT "Ideal" Ranges

- Testosterone: cisfemale range <55
- Estradiol: expert consensus recommends maintaining levels approximate to mid-cycle range for menstruating cisgender women (100-200)
  - Depending on method of administration it can be difficult to obtain labs at the ideal time

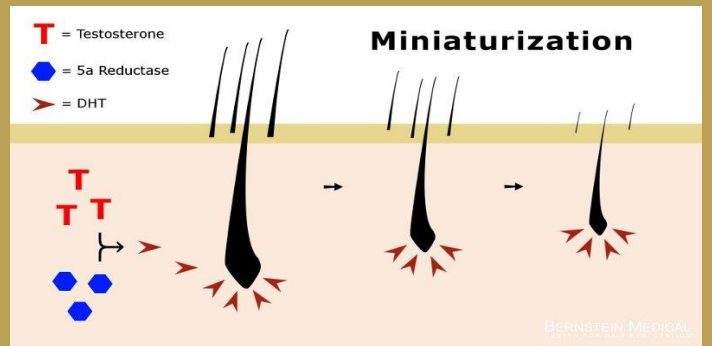
COMMON CLINICAL CONCERNS IN GENDER-AFFIRMING CARE

ANDROGENIC ALOPECIA

Androgenic alopecia (AGA) is a primary concern reported by many TGNB patients

Why does it happen?

- Testosterone is converted to DHT via the 5a reductase enzyme. DHT binds to androgen receptors in hair follicles, particularly in the frontotemporal and vertex regions, causing progressive follicular miniaturization in individuals with genetic predisposition.

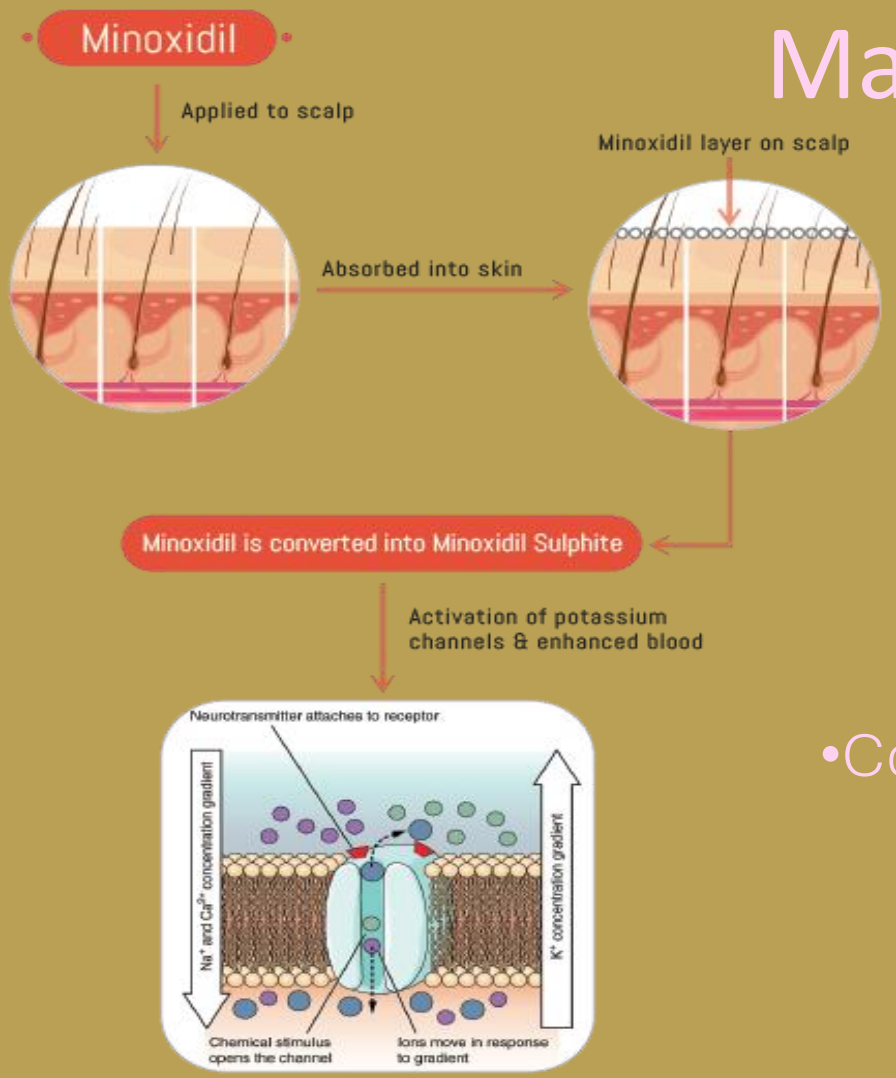


How common is it?

- Many transfeminine spectrum patients who initiate GAHT later in life will already have some degree of AGA when presenting for care
- Approximately 31-33% of transmasculine individuals develop AGA with long-term testosterone use
- Hairline shape may change within the first year on GAHT but AGA generally does not start until at least 12 months after initiation
- Severity is not clearly correlated with testosterone or DHT levels
- Genetics play a large role in AGA but are polygenic and complex (paternal and maternal lines contribute), significant family history of AGA may be predictive of future AGA potential

ANDROGENIC ALOPECIA:

Pharmacologic Management



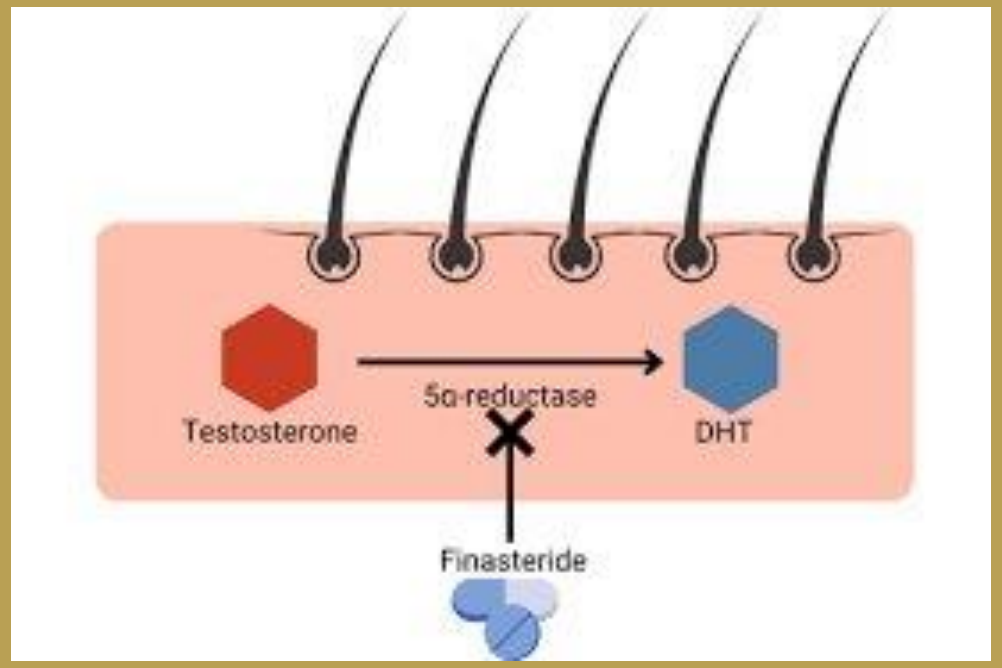
- **Minoxidil (Topical & Oral)**
  - MOA: relaxes blood vessels--> increased supply of nutrients to the hair follicles
  - PO Dose: 2.5-5mg
  - Possible SE: hypertrichosis (face/body), hypotension
  - Start w/ half dose for 2-4 weeks, if no cardiac side effects can increase to 1 tab daily
  - Counseling point: can cause hair shedding in the first month after initiation
  - Consider for transmasculine pts who experience inhibition of desired secondary sex characteristics w/ 5ARi
  - Use caution in transfeminine patients due to risk of hypertrichosis

ANDROGENIC ALOPECIA

Pharmacologic Management

5-alpha-reductase inhibitors (Finasteride, Dutasteride)

- MOA: Inhibit conversion of testosterone into DHT
- Good adjunct for transfeminine spectrum pt on estradiol monotherapy, nonbinary pts on SERM who do not desire full androgen suppression
  - Can falsely elevate total testosterone levels
- May inhibit desired androgenic secondary sex characteristics in transmasculine pts



If single agent not sufficiently effective can use combined tx +/or Topical therapies

GENITAL ATROPHY

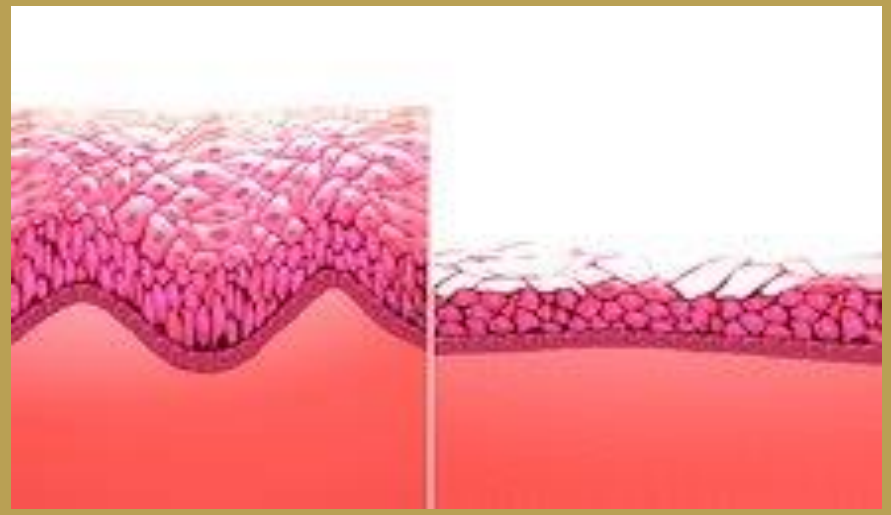
Transmasculine Spectrum

- Why?

  Exogenous testosterone suppresses HPG axis--->

  Decreased or no estrogen production---->

  Loss of vaginal epithelial cells



⇩ estrogen= ⇩ glycogen= ⇩ in lactobacilli ⟹ ⇧ vaginal pH + ⇧ vaginal infections

- Clinical Presentation
  - Vaginal dryness & irritation
    - Some people experience increase in physiologic discharge
  - Pain with penetration
  - Sometimes bleeding w/sex, itching, pain with exercise, tenderness present
  - Urinary sxs sometimes co-occur: UTIs, incontinence

GENITAL ATROPHY

Transmasculine Spectrum

Treatment

- **First line:** lubricants to increase vaginal moisture
- If no improvement… **Add back local estrogen**
  - Topical/External: estradiol 0.01% (Estrace) cream applied to vestibule daily x 2 weeks then twice weekly
  - Internal: estradiol 10mg tablets intravaginally daily x 2 weeks then twice weekly (can use cream if not covered by insurance)
- Pt Education
  - Little/no systemic absorption, will not affect progress of transition
  - Often have classically "feminine" names and packaging (i.e Vagifem)

GENITAL ATROPHY

Transfeminine Spectrum

- Why?

Exogenous estradiol suppresses HPG axis + antiandrogens directly inhibit testosterone secretion and androgen receptor binding--->

Decreased or no testosterone production & reduction of androgen stimulation---->

Decreased testicular volume and eventual testicular & penile atrophy

Atrophy can cause a diverse array of symptoms including;

- <u>Infertility</u>
  - Ensure appropriate counselling prior to GAHT initiation, assist with fertility preservation if desired
- <u>Erectile dysfunction</u>
  - Difficulty maintaining erections or having less firm erections
  - Offer PDE5 inhibitor (Tadalafil, sildenafil, ect.)

GENITAL ATROPHY

Transfeminine Spectrum

- Painful erections
  - Often leads to avoidance of erections, which increases pain when they do occur
  - Insufficient evidence to recommend topical testosterone
  - Mainstay of tx is to encourage more frequent erections in controlled environment over time
  - Rule out Peyronie's Disease
- Hematuria: fairly common amonst transwomen, particularly those who tuck
  - Assess for co-occuring symptoms: dysuria, discharge
    - If related to atrophy/tucking will generally be painless and without other symptoms
  - Rule out infectious or renal cause
    - Urinalysis w/ culture
    - STI urine

GENITAL ATROPHY

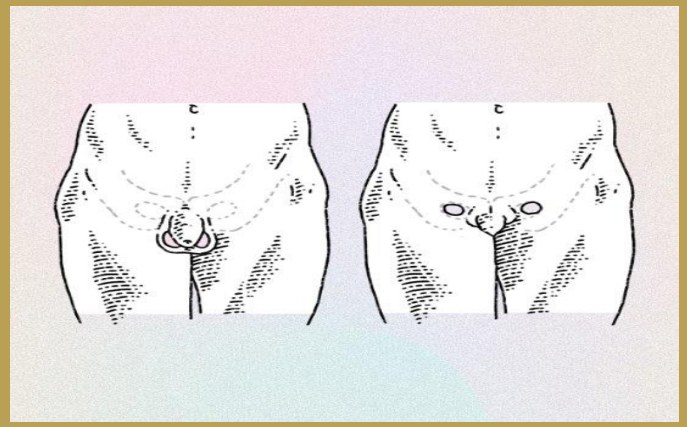
Transfeminine Spectrum

- Patient Education
  - Discuss changes in sexual desire and function in neutral manner and explore patient feelings about these changes. If patient is distressed by symptoms can discuss alternative androgen blocking strategies and offer appropriate management options
  - Be sure to include mental health and relational safety/dynamics in your assessment and recommend therapy and/or sex therapy as indicated

Advise on safer tucking strategies

DOs

- Keep skin clean and dry and monitor for s/sxs of infection
- Wear Gaffs designed for tucking or shapewear with stretch/breathability
- Take breaks & avoid tight compression



DON'Ts

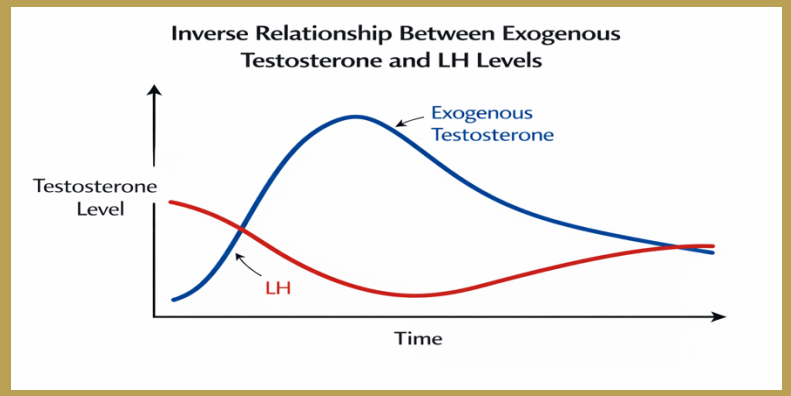
- Tuck if you have open skin, sores, or active infection
- Use duct tape or non-skin-safe adhesives.
- Tuck overnight or excessively tightly

ADDRESSING PERSISTENT BLEEDING ON TESTOSTERONE

Breakthrough blooding is common in transmasculine people on testosterone, occuring in 24-35% of individuals even after initially achieving amenorrhea

Assessment: History

- Bleeding Pattern: cyclic or random/spontaneous
- Associated symptoms: cramping? Mood changes? Dyspareunia?
- Testosterone formulation and consistency of dosing
  - Biweekly dosing is MUCH more likely to lead to breakthrough bleeding
- Sexual Activity: STIs, pregnancy risk
- Medications



Assessment: initial work-up

- Hormone levels (T/E2) and LH
  - (HPG axis suppression)
  - Thyroid function
  - STIs (if active)
- Pregnancy Test (if sperm-bearing partners)

ADDRESSING PERSISTENT BLEEDING ON TESTOSTERONE

Management: use a step-wise approach based on pt history, lab and other data

- Start with lowest-impact interventions & avoid trying multiple interventions at once
- Educate pt on the Pros and Cons of each option

- **Pharmacologic interventions to stop bleeding (short-term)**
  - **NSAIDs**
    - MOA: prostaglandin synthesis inhibitors, encourage vasoconstriction & platelet aggregation
      - Will decrease bleeding but will likely not achieve complete cessation
      - HMB decrease of 10-50%
    - Appropriate agents and dosing
      - Naproxen: 500mg BID x 5 days or Ibuprofen: 600mg qD or BID x 5 days
      - Mefenamic Acid: 500mg loading dose, then 250mg q6 hours for 7 days max
  - **Tranexamic Acid**
    - MOA: antifibrinolytic (HMB decrease of 43-63%)
    - Dosing: 1300mg TID x 5 days
    - Safe and effective

ADDRESSING PERSISTENT BLEEDING ON TESTOSTERONE

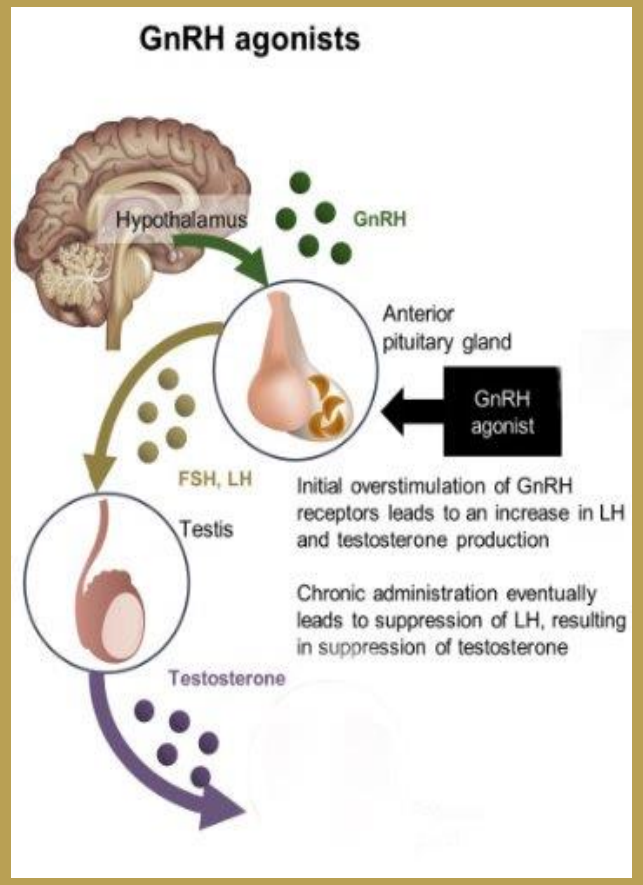
Pharmacologic interventions to stop bleeding (long-term)

Hormonal Contraception

- 1st line: COCs: **Continuous use** (skip placebo week)
  - More effective than progestogens alone
- Progesterone-only methods: IUD>Nexplanon> Depo-provera.
  - Likely to cause irregular bleeding patterns

GnRH agonists (i.e Lupron)

- When given continuously they act as an inhibitor of the pituitary gonadotropins LH and FSH & cause downregulation of GnRH receptors on the pituitary gland = complete HPG axis suppression
- Dosing: 11.5mg monthly or 22.5mg q3-mo
- Administered in-clinic

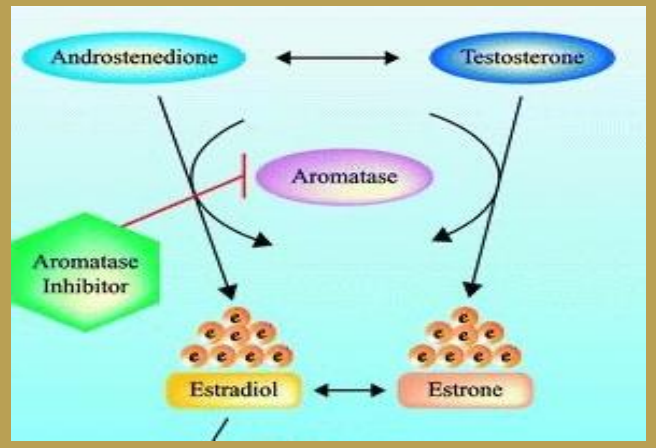


ADDRESSING PERSISTENT BLEEDING ON TESTOSTERONE

Pharmacologic interventions to stop bleeding (long-term)

Danazol

- Chemically derived from testosterone. Inhibits ovulation and reduces estrogen levels. Causes endometrial atrophy, which leads to amenorrhea
- Also used to treat endometriosis



Aromatase Inhibitors

- MOA: Inhibits conversion of testosterone into estradiol
- Preferred agent: Anastrozole-more selective & potent
- Maybe of particular benefit for patients with significant adipose tissue as aromatase is stored in adipose tissues
- Generally well tolerated; menopausal-like side effects possible (fatigue, headache, hot flashes, vaginal dryness)

ADDRESSING PERSISTENT BLEEDING ON TESTOSTERONE

Inquire about consistency of dosing → Adjust dose/frequency of testosterone → Address other contributing factors → Pharmacologic management

If initial assessment is unrevealing, labs are within expected range and interventions are not effective to stop bleeding, consider TVUS to r/o structural pathology

REFERENCES

Coleman, E., Radix, A. E., Bouman, W. P., Brown, G. R., de Vries, A. L., Deutsch, M. B., Ettner, R., Fraser, L., Goodman, M., Green, J., Hancock, A. B., Johnson, T. W., Karasic, D. H., Knudson, G. A., Leibowitz, S. F., Meyer-Bahlburg, H. F., Monstrey, S. J., Motmans, J., Nahata, L., … Arcelus, J. (2022). Standards of care for the health of transgender and gender diverse people, version 8. *International Journal of Transgender Health*, *23*(sup1). https://doi.org/10.1080/26895269.2022.2100644

DeLoughery, Thomas. "Hematologic Concerns in Transgender Patients." *Clinical Advances in Hematology & Oncology*, vol. 20, no. 8, Aug. 2022, pp. 516–523, www.hematologyandoncology.net/files/2022/07/ho0822_deloughery_feature_v1-1.pdf. Accessed 24 Aug. 2024.

Grimstad FW, Boskey ER, Clark RS, Ferrando CA. Incidence of breakthrough bleeding in transgender and gender-diverse individuals on long-term testosterone. Am J Obstet Gynecol. 2024 Apr;230(4):434.e1-434.e10. doi: 10.1016/j.ajog.2023.12.036. Epub 2024 Jan 4. PMID: 38181830.

Grimstad FW, Boskey ER, Clark RS, Ferrando CA. The Journal of Sexual Medicine. 2023;20(12):1459-1465. doi:10.1093/jsxmed/qdad135.

Guidelines for the primary and gender-affirming care of transgender and gender nonbinary people. Guidelines for the Primary and Gender-Affirming Care of Transgender and Gender Nonbinary People | Gender Affirming Health Program. (2016, June 17). Retrieved March 14, 2023, from https://transcare.ucsf.edu/guidelines

Hembree, W. C., Cohen-Kettenis, P. T., Gooren, L., Hannema, S. E., Meyer, W. J., Murad, M. H., Rosenthal, S. M., Safer, J. D., Tangpricha, V., & T'Sjoen, G. G. (2017). Endocrine treatment of gender-dysphoric/gender-incongruent persons: An endocrine society* clinical practice guideline. The Journal of Clinical Endocrinology & Metabolism, 102(11), 3869–3903. https://doi.org/10.1210/jc.2017-01658

Milionis C, Ilias I, Koukkou E. Liver function in transgender persons: Challenges in the COVID-19 era. World J Clin Cases. 2023 Jan 16;11(2):299-307. doi: 10.12998/wjcc.v11.i2.299. PMID: 36686345; PMCID: PMC9850971.

Raven, L.M., Guttman-Jones, M. & Muir, C.A. Hyperprolactinemia and association with prolactinoma in transwomen receiving gender affirming hormone treatment. *Endocrine* 72, 524–528 (2021). https://doi.org/10.1007/s12020-020-02563-3

Wellington K, Wagstaff AJ. Tranexamic acid: a review of its use in the management of menorrhagia. Drugs. 2003;63(13):1417-33. doi: 10.2165/00003495-200363130-00008. PMID: 12825966

Wiepjes, C. M., Nota, N. M., Adank, M. A., A Dreijerink, K. M., Barbé, E., & M Konings, R. H. (2019). Breast cancer risk in transgender people receiving hormone treatment: Nationwide cohort study in the Netherlands. *The BMJ*, *365*. https://doi.org/10.1136/bmj.l1652